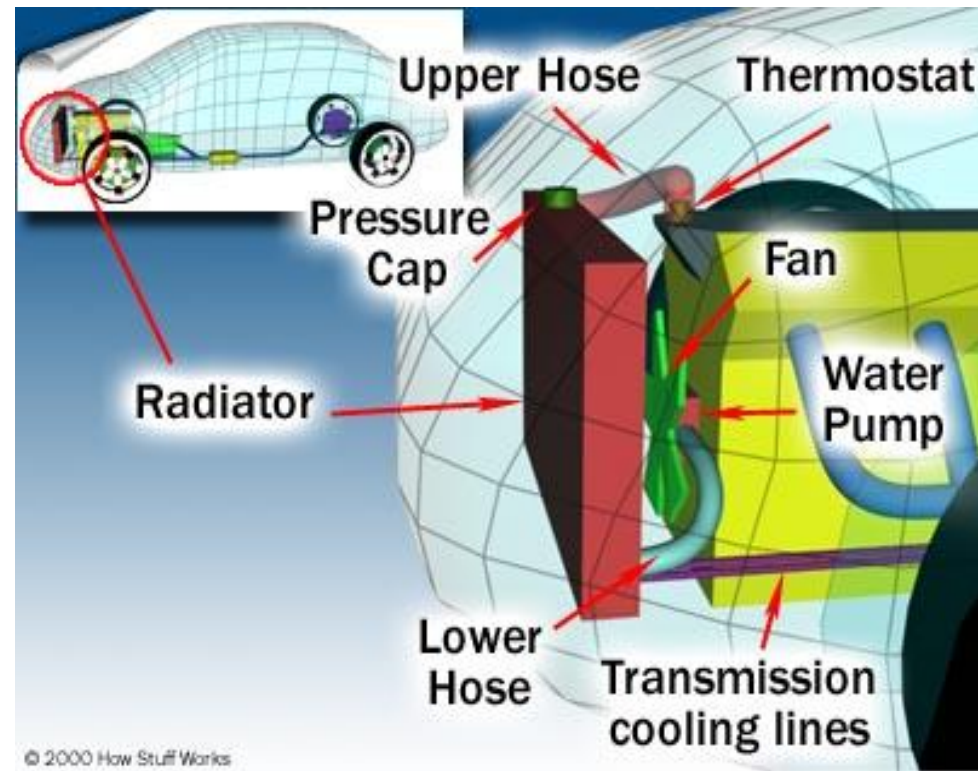


Chapter 7 Cooling System

发动机冷却系

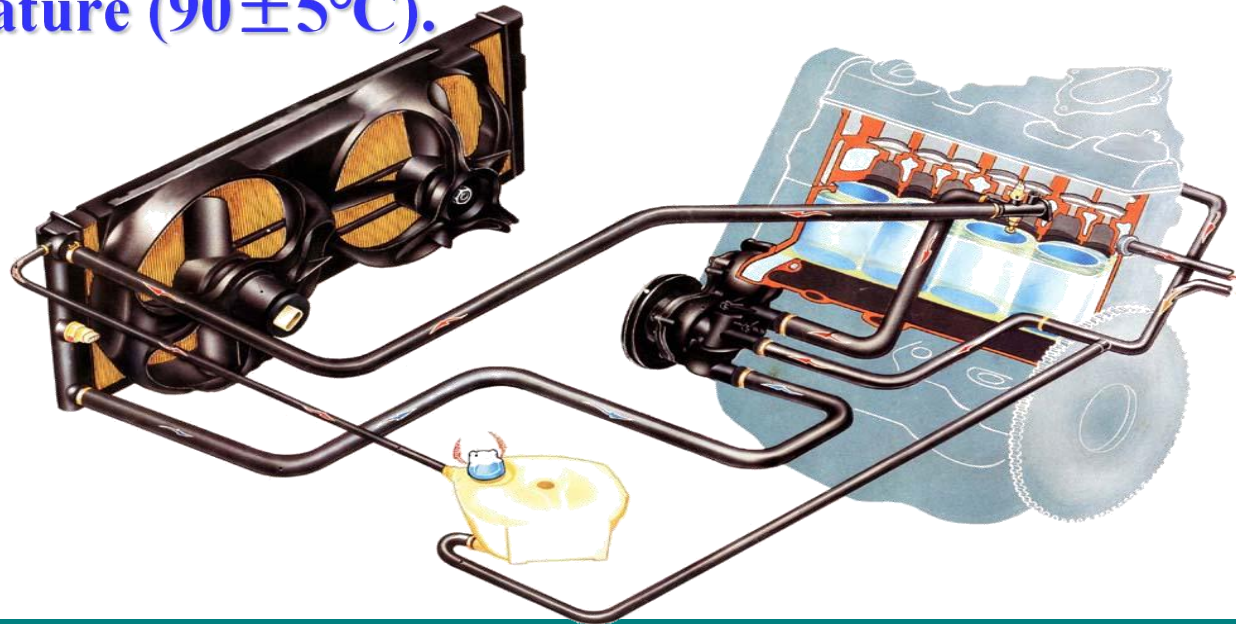
- 7.1 Components of Cooling System (冷却系组成)
- 7.2 Coolant (冷却液)
- 7.3 Radiator (散热器)
- 7.4 Radiator Fan (冷却风扇)
- 7.5 Thermostat (节温器)
- 7.6 Water Pump (水泵)
- 7.7 Oil Cooling (机油冷却)



7.1 Components of Cooling System (冷却系组成)

The growing demands with respect to fuel consumption, exhaust gas emissions, service life, ride comfort, and package have led to modern cooling systems for internal combustion engines in the motor vehicles.

Cooling system is used to keep the engine at proper temperature ($90 \pm 5^\circ\text{C}$).

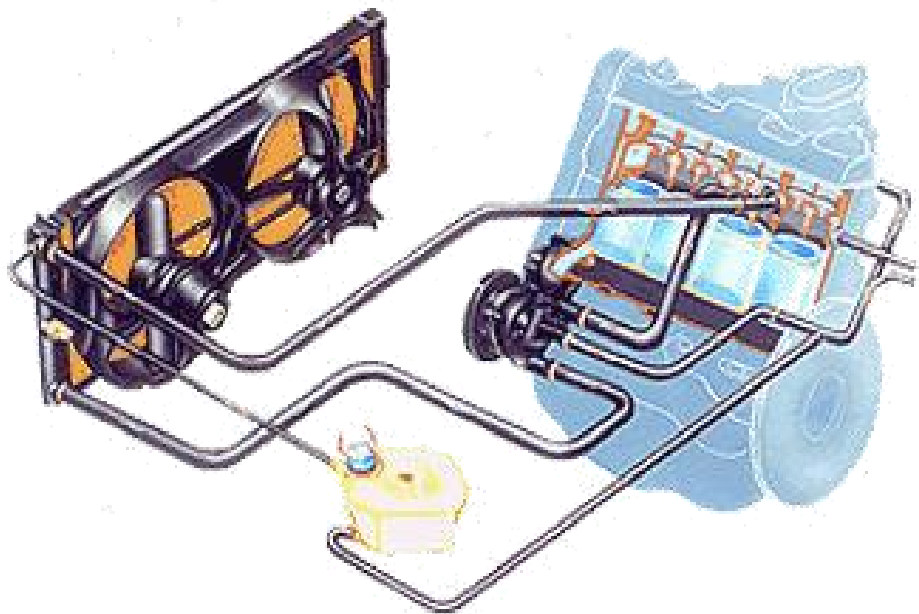


Influences of cooling degree

Cooling degree	Effects
Too cold	(1)More heat loss (2)For diesel engine, oil viscosity (粘度) increases which results in higher friction loss (摩擦损失) ; for gasoline engine, gasoline condenses in cylinder and flows into crankcase, influencing lubrication(润滑). (3)Mixture is difficult to be ignited (4)Increased oil viscosity results in serious wear
Too hot	(1)Lower volumetric efficiency(充气效率); lower power (2)Easier to deflagrate(爆燃) (3)Normal gap between moving components is destroyed which causes serious wear ; (4)Worse lubrication (5)Worse mechanical properties of parts causes damage

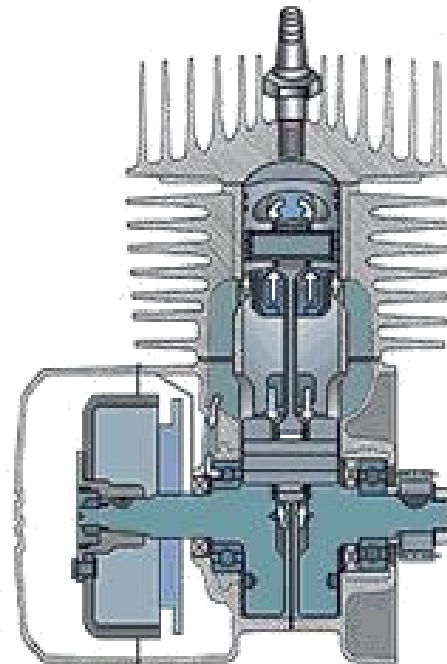
Classification of cooling system (冷却系分类)

Water cooling



水冷

Air cooling



风冷

Temperature of engine at proper cooling conditions

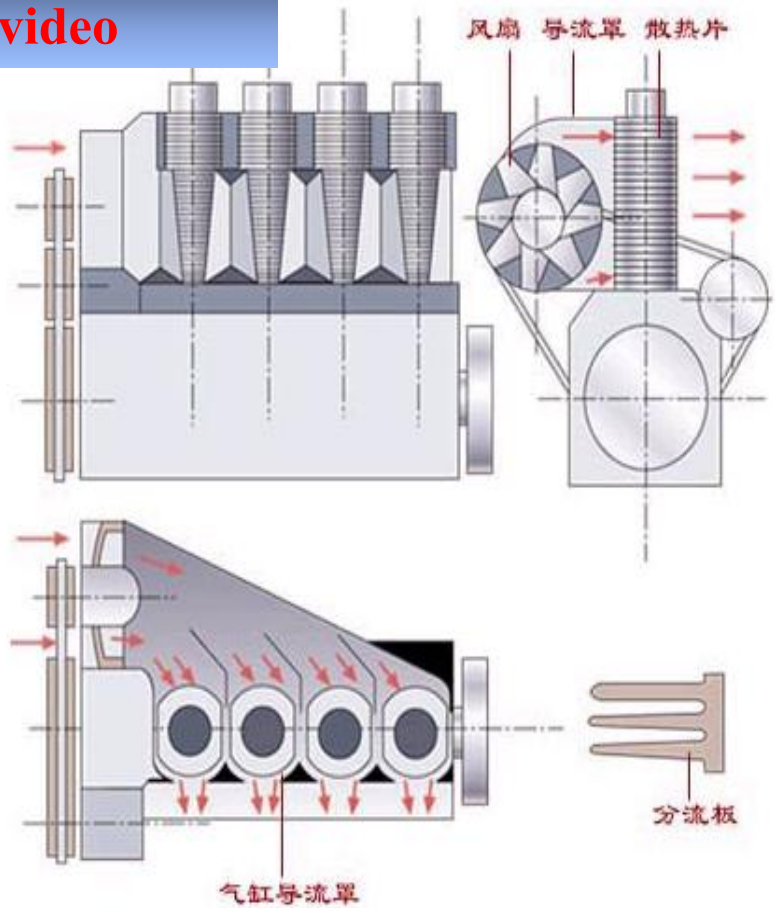
Systems	Temperature range(温度范围)
Water cooling	Coolant in cylinder head: 353~363K Cylinder wall(气缸壁): below 470~550K
Air cooling	Cylinder head: 423~453K Cylinder wall: 433~473K

Comparison between these two cooling systems

Systems	Advantages and disadvantages	Application
Water cooling	Powerful cooling Easy to regulate Easy to start in winter	Widely used in automotive engines
Air cooling	Bad cooling Big noise Big power loss	Small displacement engines and military vehicle engines (军车发动机)

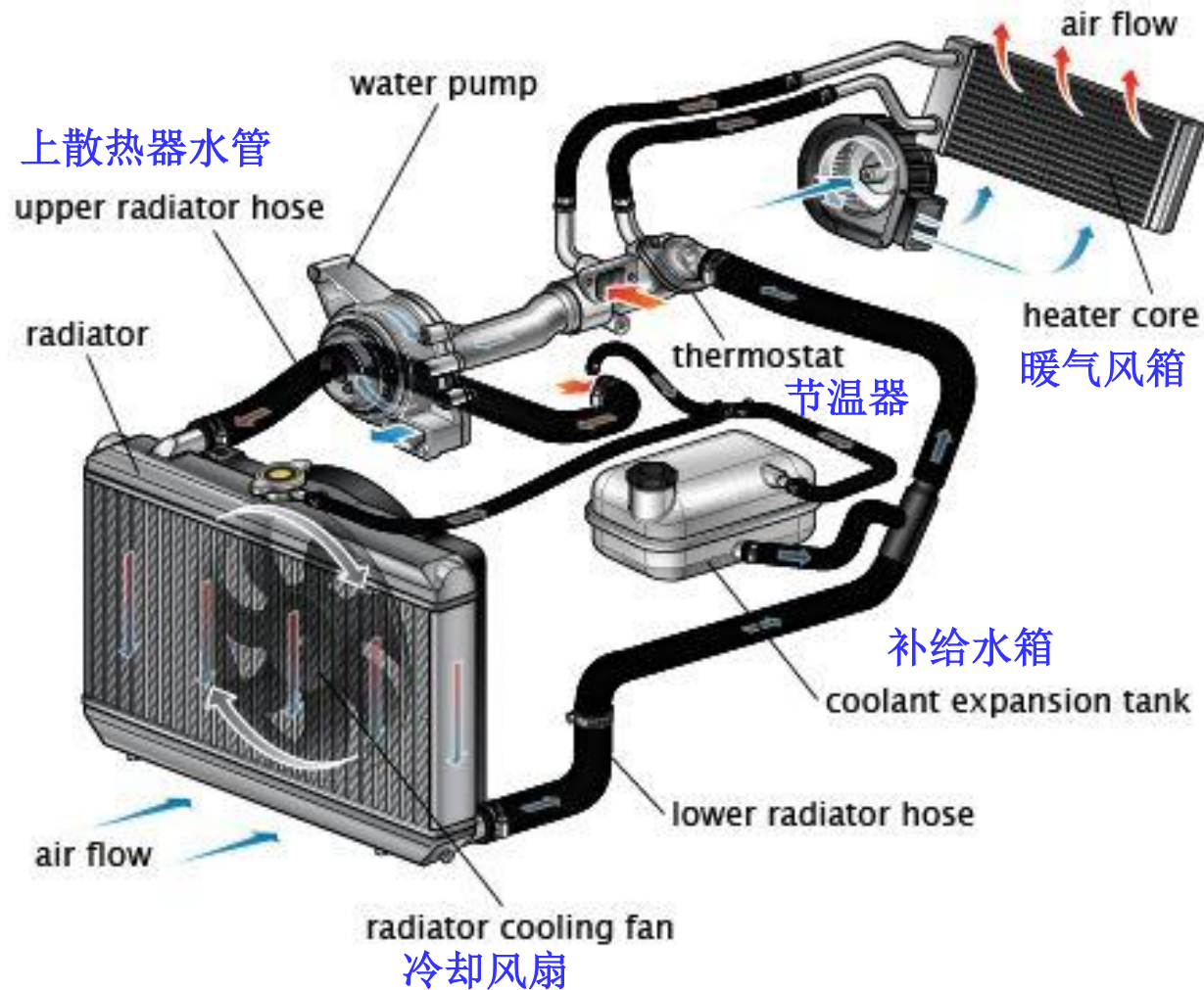
Air cooling system

video



Air cooling system takes use of the rapid flow of air to take the heat from cylinder wall .

Components of water cooling system



Basic components of water cooling system are coolant, radiator, radiator fan, water pump, thermostat and different hoses.

7.2 Coolant(冷却液)

Coolant performs two critical functions. It keeps the radiator fluid from freezing in wintry conditions and keeps the engine from overheating.

The main composition of coolant is water because of its huge specific heat capacity(比热容) and it is easy to get . However, if the water contains part of calcium(钙), magnesium(镁) and other mineral salts(矿物盐类) which will form scaling(结垢) in engines, the proper cooling process will be damaged. And electrochemical reaction erodes(电化学反应腐蚀) cooling system. So without any additive(添加剂), we should only use soft water(软水) as coolant.

However , pure water freezes at 0°C which cannot meet the need. So we usually add antifreeze(防冻剂) in water to raise its boiling point and lower its freezing point(冰点).

Coolant contains: water/ antifreeze/additive



The mixture of ethylene glycol(乙二醇) and water is widely used.

Freezing temperature /°C	ethylene glycol (%volume)	water (%volume)	density g/cm ³
-10	28.4	71.6	1.043
-15	32.8	67.2	1.0426
-20	38.5	61.5	1.0506
-25	45.3	54.7	1.0586
-30	47.8	52.2	1.0627
-40	54.7	45.3	1.0713
-50	59.9	40.1	1.0780

There are different kinds of antifreeze, which are most easily identified by their color.

“The green stuff” is traditional coolant, which can be used in most cars. It contains lubricating silicates(润滑硅酸盐) and corrosion inhibitors(缓蚀剂), but these silicates deteriorate(恶化) rather quickly, requiring coolant changes every two years/24,000 miles.

Coolant can also be red, yellow, orange, or even purple. These coolants are pretty much the same; they are simply dyed (染色) different colors. They have a longer claimed service life, thus requiring fewer fluid changes-in some cases up to 100,000 miles.

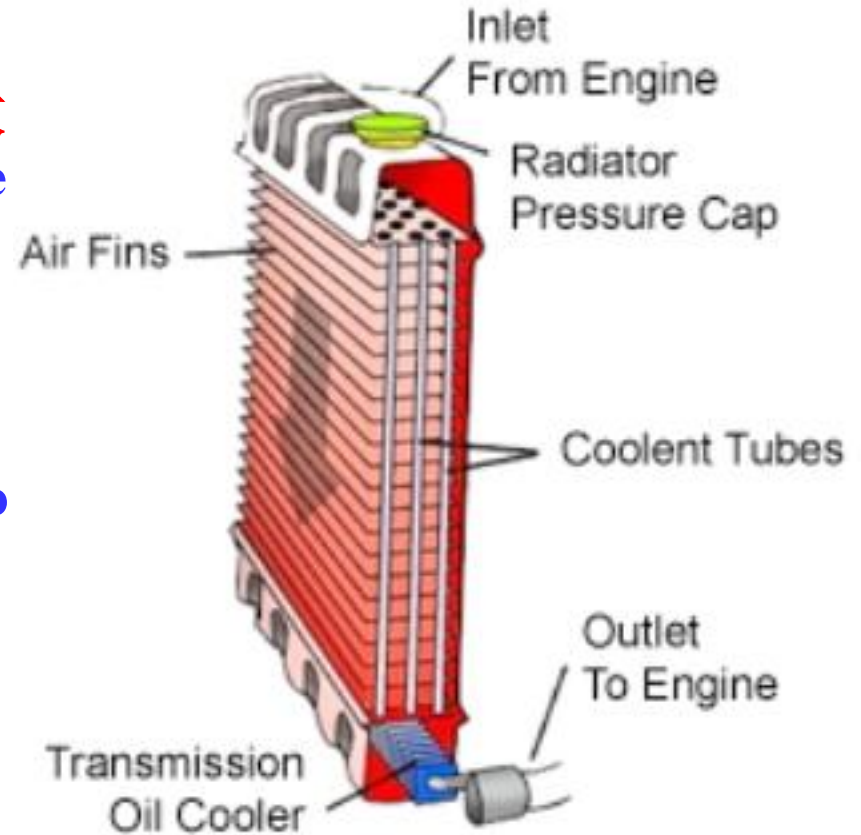


Red and Green Engine Coolant

7.3 Radiator (散热器)

The radiator is a heat exchanger with hundreds of individual tubes and fins(散热器片) that reduce the temperature of the coolant.

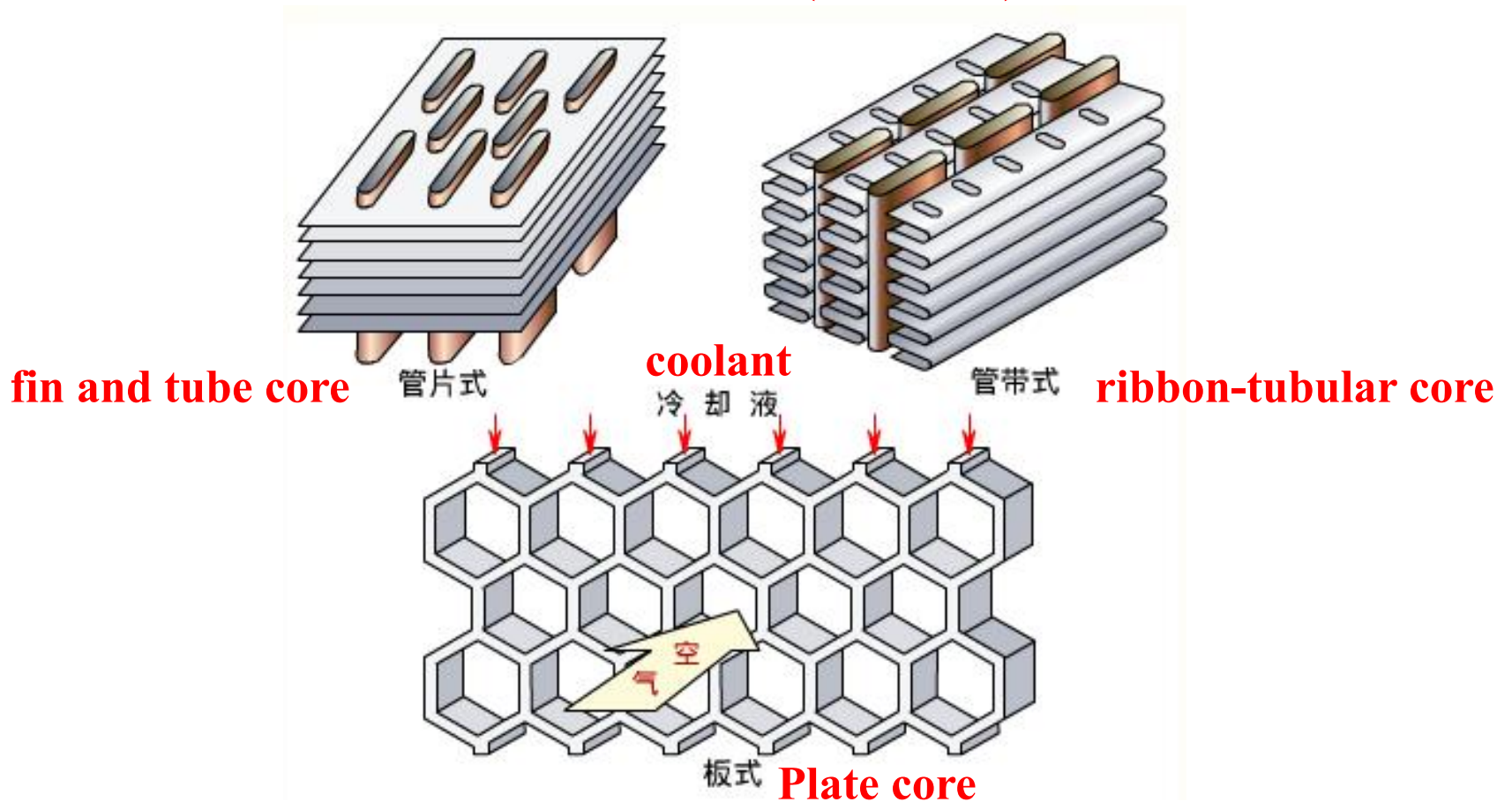
As coolant travels through the water jacket (水套) it absorbs and removes heat from the engine and transports it to the radiator. Air flows through the coolant passages as the car moves, cooling the tubes and fins and coolant reenters the engine with a reduced temperature.



散热器材料：黄铜、铝或复合材料

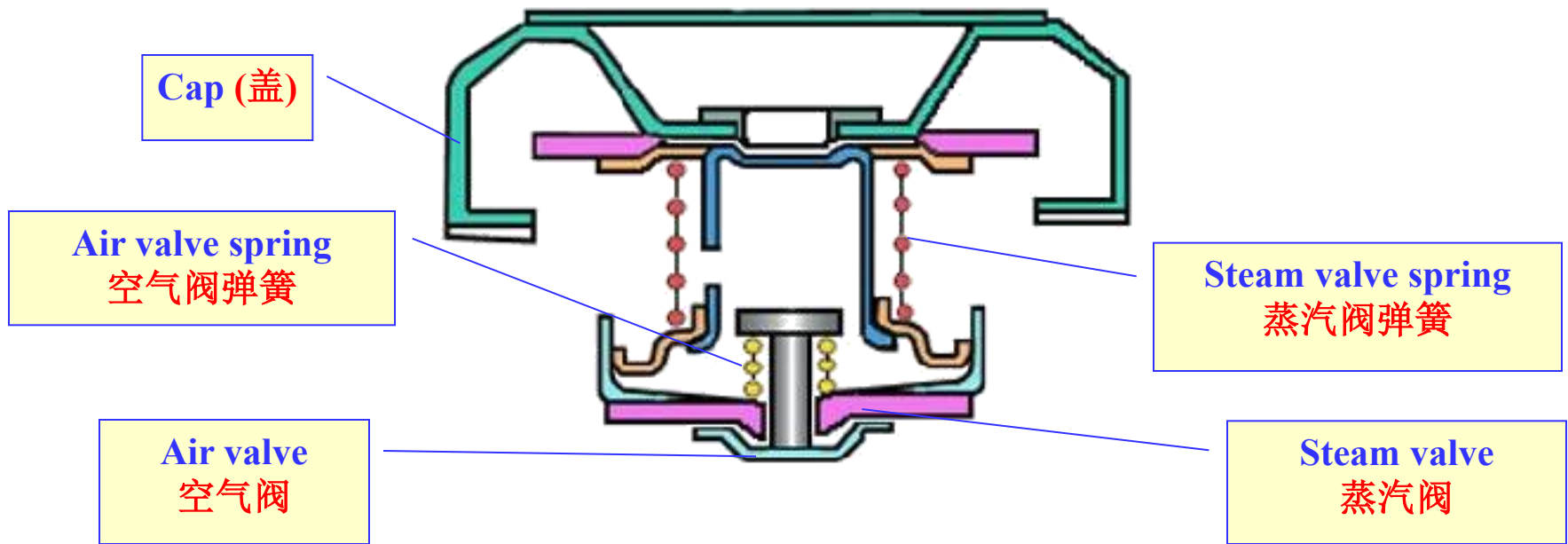
Radiator is made of brass, aluminum or composite materials.

Radiator Core (散热器芯)



Radiator cap(散热器盖)

The radiator cap is a pressure-release valve. It keeps the cooling system under pressure which is higher than ambient pressure in order to increase the boiling point of coolant.

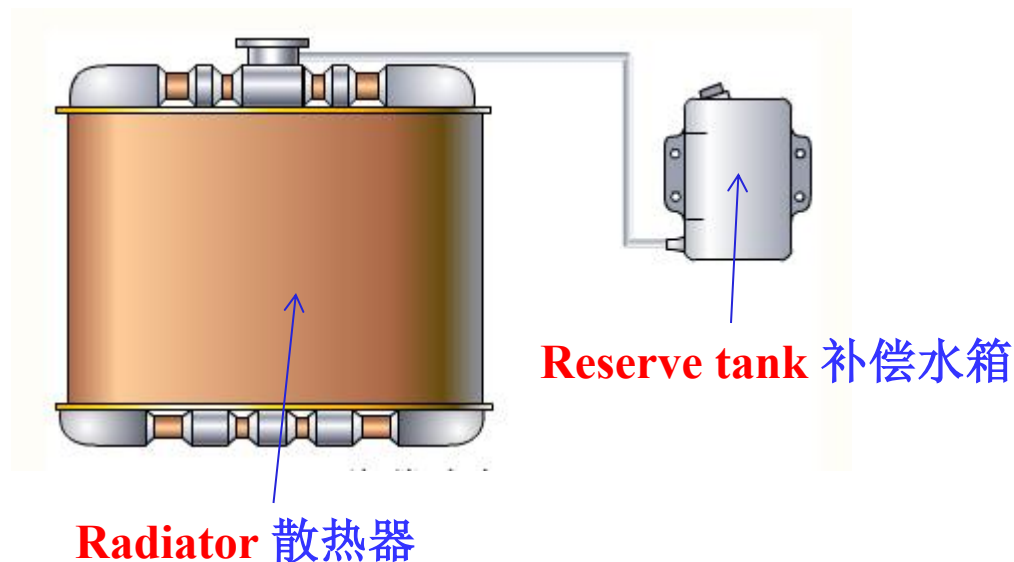


Working process of radiator cap



Reserve tank(补偿水箱)

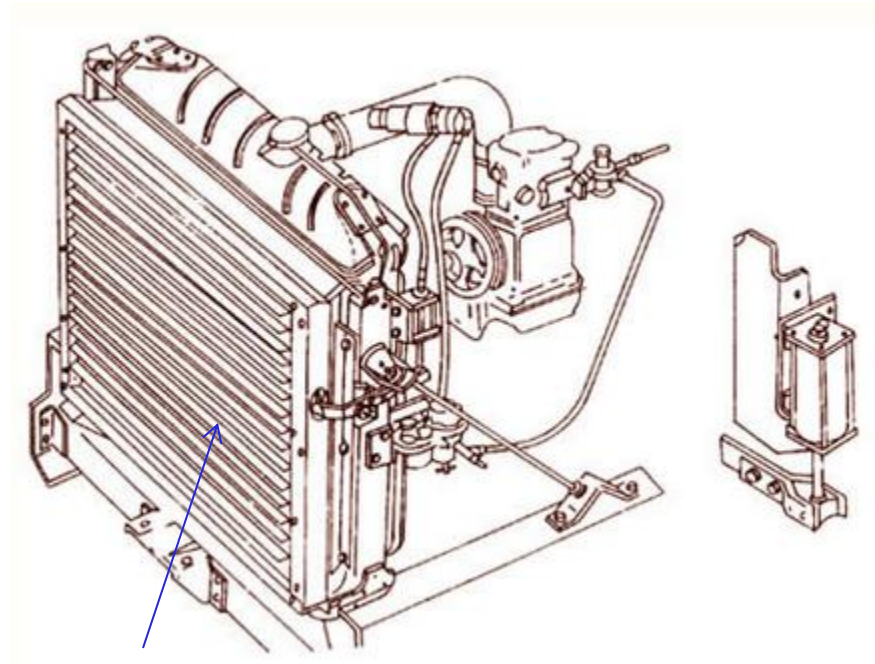
When the cooling liquid expands when heated , part of the coolant flows into the reserve tank; when the cooling liquid is cooling , the part of the coolant is sucked back into the radiator. As a result , the coolant won't overflow from radiator and it keeps the radiator full of coolant.



其作用就是当冷却液受热膨胀时，部分冷却液流入补偿水桶；而当冷却液降温时，部分冷却液又被吸回散热器，冷却液不会溢失。

Shutter(百叶窗)

To control the opening state of shutter, we can regulate the mass of air flowing to the radiator. So the degree of cooling can be controlled.

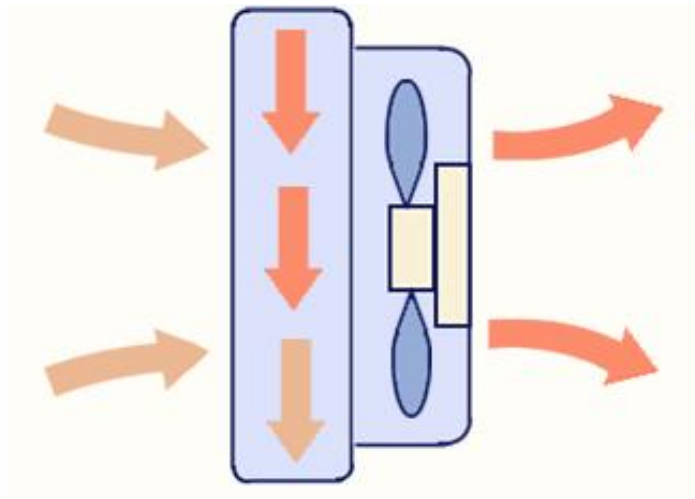


Shutter

功用：通过调节流经散热器的空气量来调节冷却系的冷却强度。

7.4 Radiator Fan (冷却风扇)

The fan is used for increasing the velocity and mass of air flowing to the radiator. It is mounted(安装) behind the radiator.



Axial flow fan(轴流式)

Category: Axial flow fan(轴流式)

Centrifugal fan(离心式)

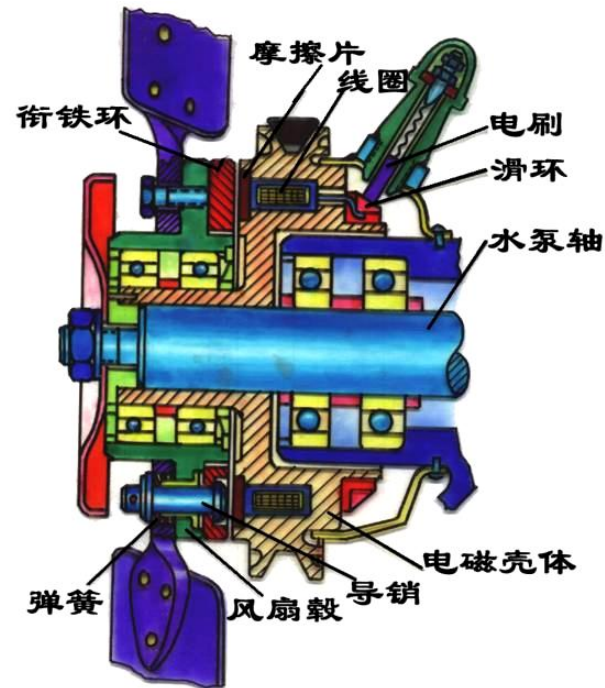
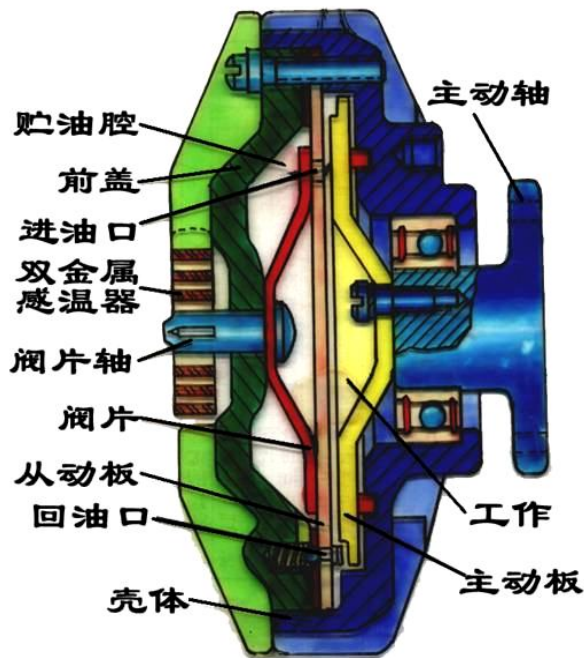
For the axial flow fan's structure is simple and it has high efficiency, so it's widely used on automotive.

The radiator fan can be driven by a drive belt or an electric motor.

Fan clutch (风扇离合器)

For the engine is not always working at high temperature, so we take use of the fan clutch to control the state of radiator fan, which will reduce some power loss.

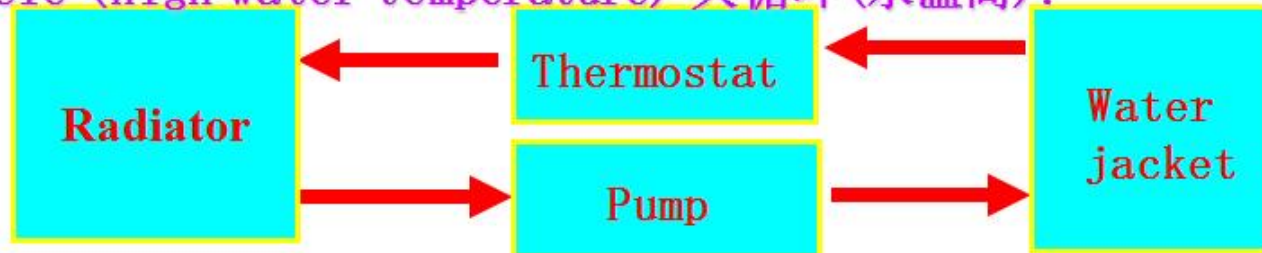
The clutch mainly contains silicone oil clutch(硅油离合器) and electromagnetic clutch(电磁式离合器).



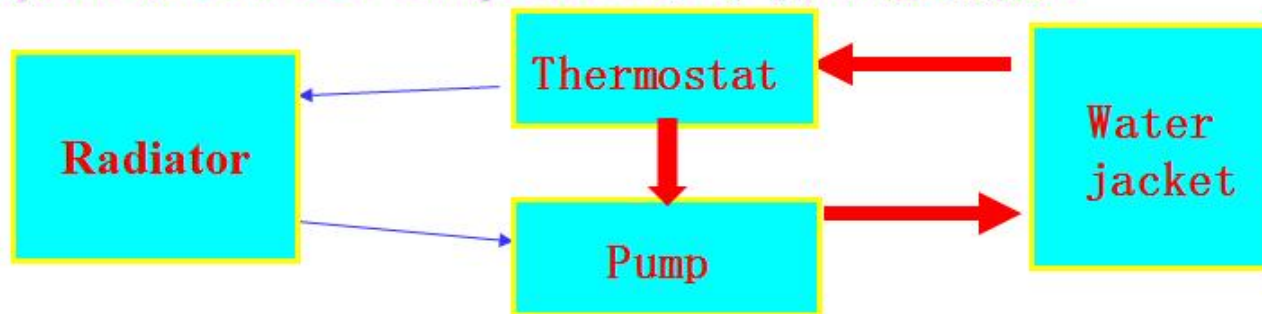
7.5 Thermostat (节温器)

The thermostat helps the engine reach operating temperature by preventing coolant from circulating to the radiator, thus allowing the engine to heat up more quickly. As the engine reaches operating temperature, the thermostat opens and allows coolant to flow through the radiator.

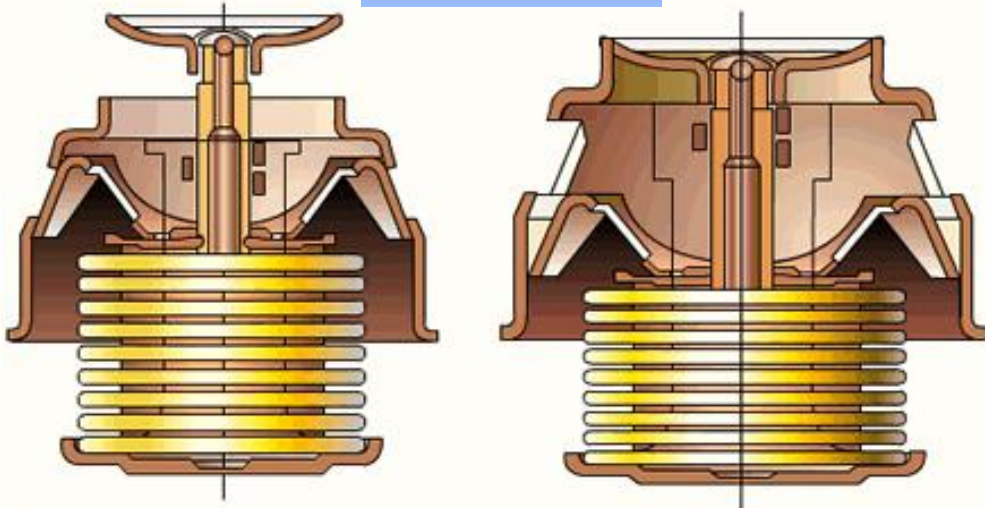
Big cycle (high water temperature) 大循环(水温高): $>89^{\circ}\text{C}$



Small cycle (low water temperature) 小循环(水温低): $<76^{\circ}\text{C}$

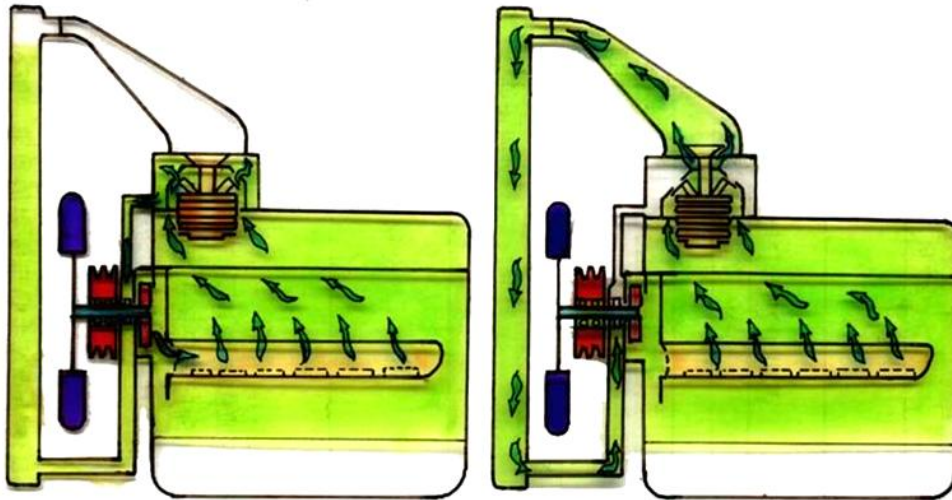


video



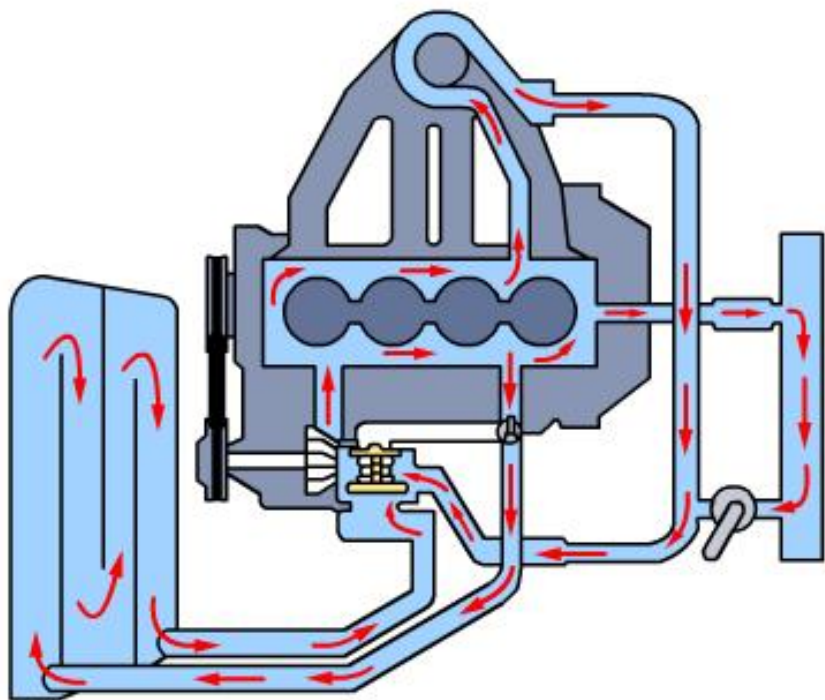
Wax-type thermostat (蜡式节温器) takes use of the feature of paraffin(石蜡) that it extends after heated from solid to liquid and pushes the spring to switch the water passage.

Thermostat using an electric heating of wax element has been created. The thermostat position can be set independently of the current coolant temperature on the basis of set point from an engine map.

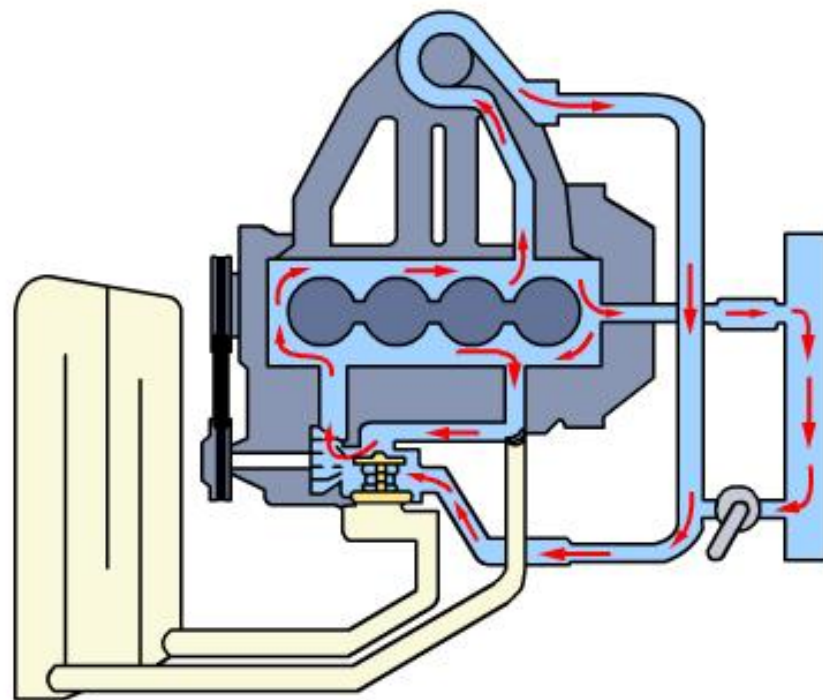


Close (small cycle)

Open (big cycle)

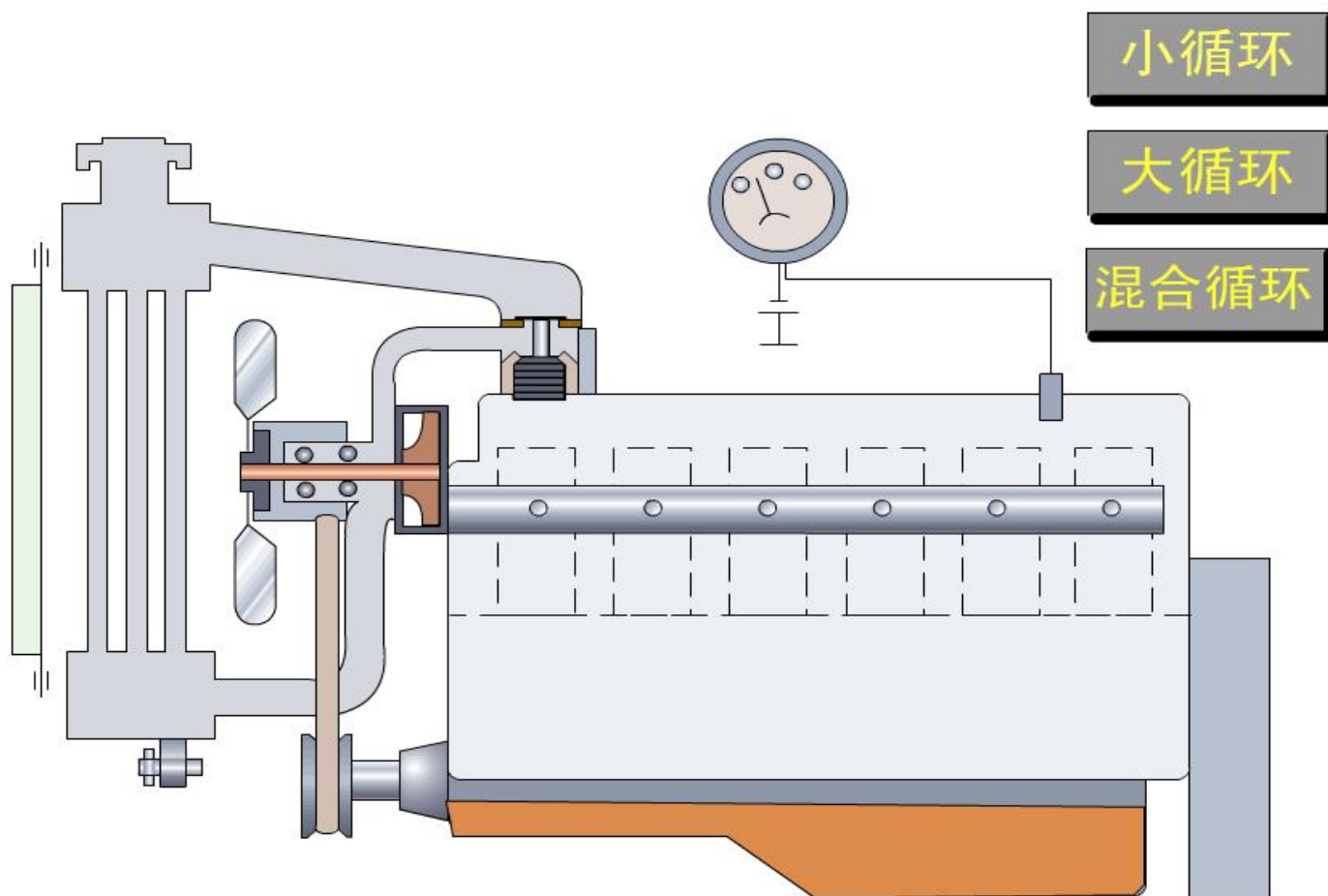


大 循 环



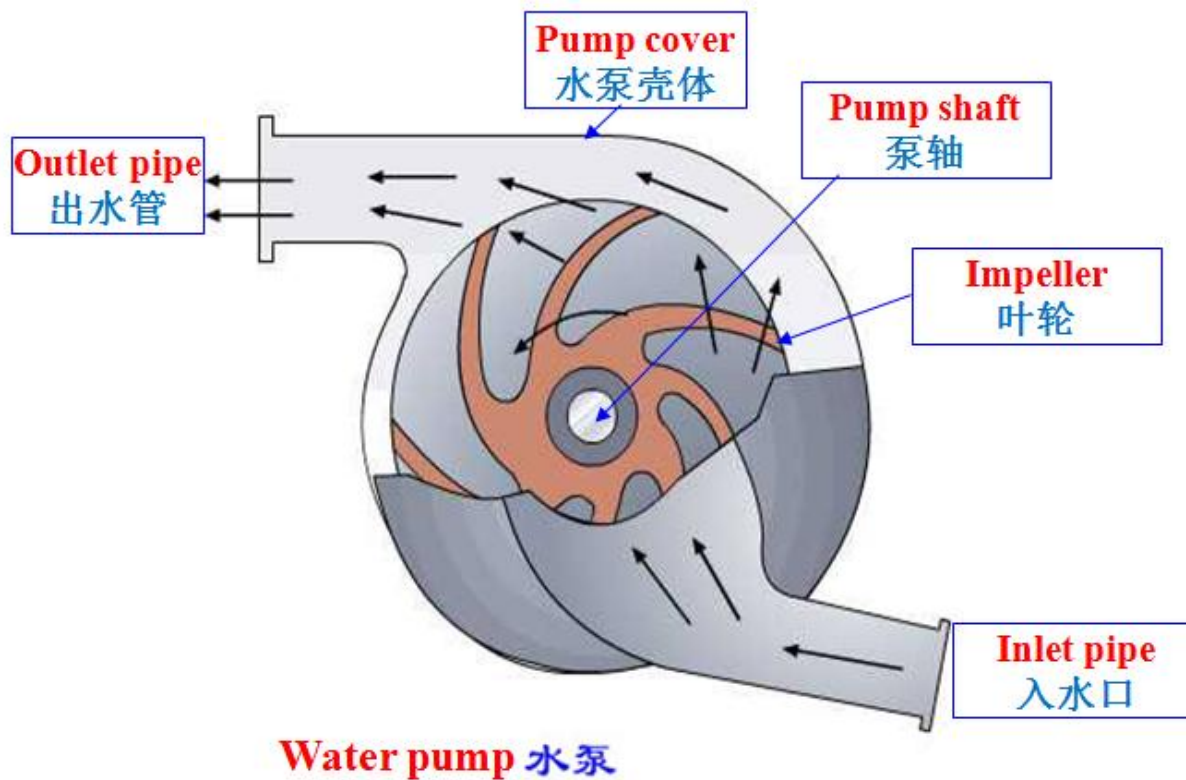
小 循 环

Schematic water cooling cycles

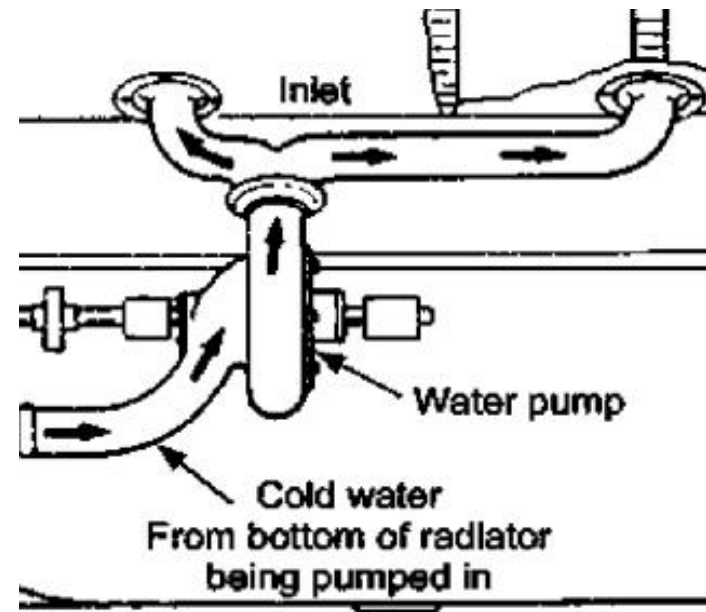
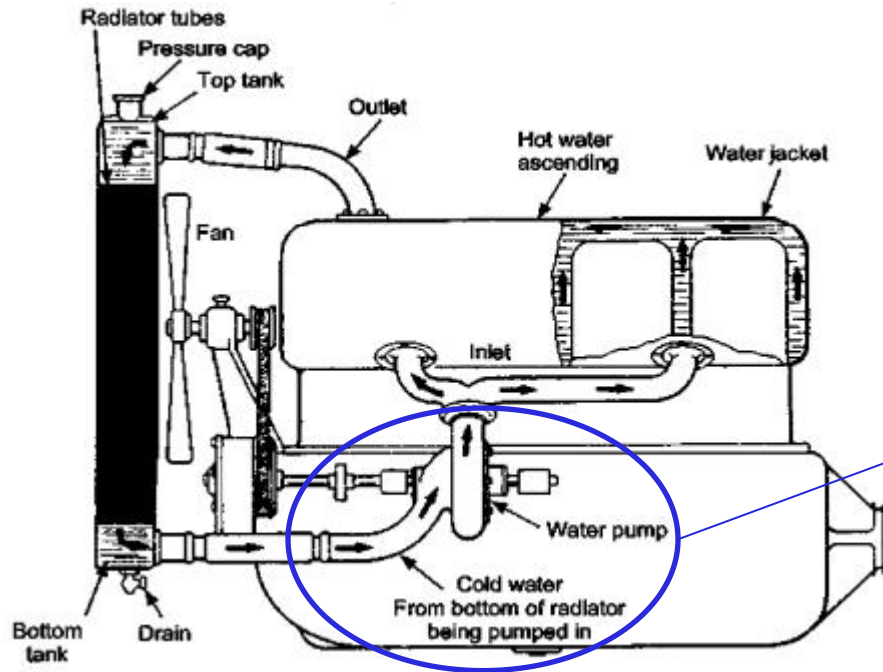


7.6 Water Pump (水泵)

The water pump circulates coolant through the system via an impeller (a rotor that spins to move fluid).



Centrifugal water pump (离心式水泵) is widely used.



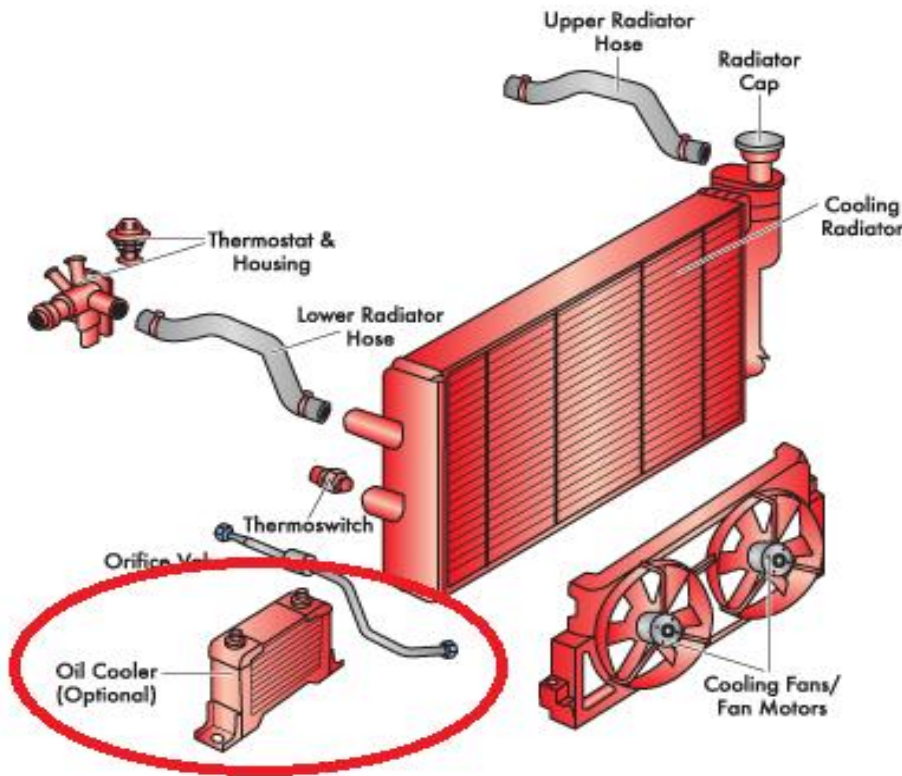
When the impeller is rotating, the water in pump rotates with the impeller and is driven to the edge of the impeller under the action of centrifugal force (离心力). So the water is pumped into water jacket leading to the formation of low-pressure area in the center of pump. Then the cold water from radiator is sucked into the pump.

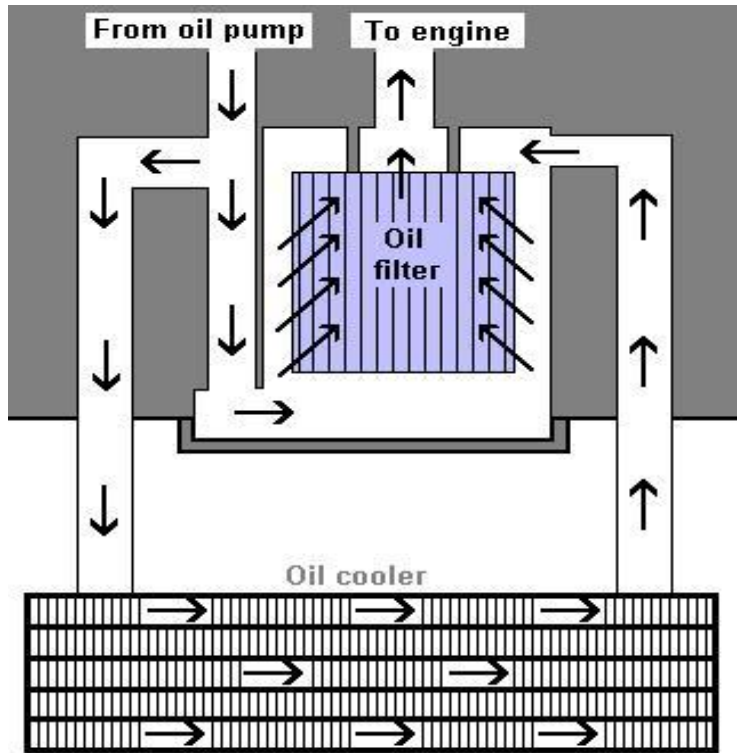
7.7 Oil Cooling (机油冷却)

Part of the heat from engine is absorbed by the lubricating oil.
With some powerful engines, temperature of oil could be too high.
So the oil cooler has to be employed.

In addition, to keep the engine cool, on cars with an automatic transmission(自动变速器), the radiator is equipped with a separate heat exchanger to keep transmission fluid from boiling over.

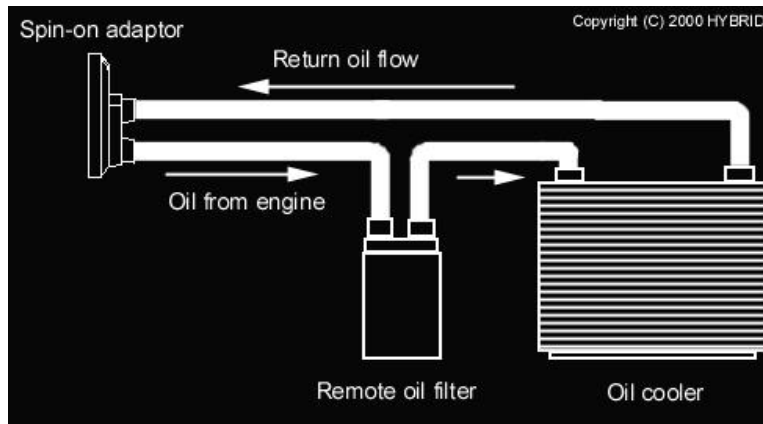
Coolant is circulated through the oil cooler to absorb the heat from the lubricating oil.





Heat is transferred to the lubricating oil as it passes around engine components such as bearings and pistons. Oil cooler is used to dissipate heat from the oil to maintain optimum temperatures for proper lubrication.

It is actually a heat exchanger. Heat from hot oil flowing through the oil cooler is absorbed by the cold water from radiator. Then the cooled oil is driven into engine.



Group homework 4

Development and Key Technologies for the New Vehicle Powertrains

车用新型动力系统的进展及关键技术特征